

EFFECTS OF HIGH-INTENSITY TRAINING BY HEART RATE OR POWER IN WELL-TRAINED CYCLISTS

JEROEN SWART, ROBERT P. LAMBERTS, WAYNE DERMAN, AND MICHAEL I. LAMBERT

UCT/MRC Research Unit for Exercise Science and Sports Medicine, Department of Human Biology, Faculty of Health Sciences, University of Cape Town, The Sport Science Institute of South Africa, Newlands, South Africa

ABSTRACT

Swart, J, Lamberts, RP, Derman, W, and Lambert, MI. Effects of high-intensity training by heart rate or power in well-trained cyclists. *J Strength Cond Res* 23(2): 619–625, 2009—The aim of this study was to determine whether the performance of cyclists after 4 weeks of high-intensity training improved similarly using either heart rate or power to prescribe training. Twenty-one well-trained men cyclists (age, 32 ± 6 years; peak power output, 371 ± 46 W) were randomly assigned to a power-based (G_{POWER}) or heart rate-based (G_{HEART}) high-intensity training (HIT) group or a control group (G_{CONTROL}). Training consisted of 8 repetitions of 4 minutes at either 80% of peak power output (G_{POWER}) or at the heart rate coinciding with 80% of peak power output (G_{HEART}), with rest periods of 90 seconds. A 40-km time trial and $\dot{V}O_2\text{max}$ test were performed before and after 8 training sessions. There were significant improvements ($p < 0.05$) in peak power output ($G_{\text{POWER}} = 3.5\%$; $G_{\text{HEART}} = 5.0\%$) and 40-km time trial performance ($G_{\text{POWER}} = 2.3\%$; $G_{\text{HEART}} = 2.1\%$) for both of the high-intensity groups. Although there were no significant differences between groups for these variables, when the data were analyzed using magnitude-based effects, the G_{HEART} group showed greater probability of a “beneficial” effect for peak power output. The current general perception that prescribing training based only on power is more effective than prescribing training based on heart rate was not supported by the data from this study. Coaches who are unable to monitor progress frequently should prescribe training based on heart rate, when intervals are performed under stable conditions, because this may provide an additional advantage over prescribing training using power.

KEY WORDS cyclists, $\dot{V}O_2\text{max}$, 40-km time trial, high-intensity training, peak power output, performance

INTRODUCTION

It is well known that endurance training results in performance improvements in previously sedentary and recreationally active individuals (1). Furthermore, it is known that athletes who are already trained need to incorporate high-intensity interval training into their training programs to improve their performance in preparation for competition (12,17). A number of authors have investigated the effects of high-intensity training (HIT) on athletic performance (16,27,28). These studies demonstrate that high-intensity, short-duration interval training can result in further increases in peak power output (W_{max}) and $\dot{V}O_2\text{max}$ in well-trained athletes.

Although the importance of HIT is widely accepted, the method of applying the training load varies. For example, there are 2 commonly used methods for applying workload during high-intensity cycle training. These are based on prescribing training intensity either according to heart rate or power output.

During training, a number of factors can alter the submaximal heart rate/workload relationship (15). However, a longitudinal study of 13 professional road cyclists found that under controlled conditions, the heart rate coinciding with various physiological markers was relatively constant (2–3 bpm) during the course of a season. In contrast, power output, which may be a more direct measure of workload, varied significantly (19).

Despite the popularity and accuracy of using power meters to monitor performance during training and racing, no studies have validated their use to prescribe training. Specifically, the use of power in preference to heart rate to prescribe training has not been justified. Accordingly, the aim of this study was to compare the training adaptations and performance characteristics of well-trained cyclists after 4 weeks of HIT using either heart rate or power output to prescribe the training intensity. We used a 40-km time trial and $\dot{V}O_2\text{max}$ test as measures of performance (21). We hypothesized that there would be no measurable differences between the 2 modes of training.

METHODS

Experimental Approach to the Problem

The minimal sample size for this study was determined using the 40-km time trial data of Palmer et al. (20). Assuming that

Address correspondence to Jeroen Swart, jeroen.swart@uct.ac.za.

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the smallest meaningful difference is 1.0% with an SD of 0.5%, the sample size required for this study to achieve a statistical power of 80% and a significance level of 5% was $n = 5$ for each group (3). The study was conducted during 18 months. Inclusion criteria for the study were a minimum of 6 training hours per week during the 6-week period before the trial and a minimum competitive cycling background of at least 3 years. All participating cyclists were randomly assigned to 1 of 3 groups: a power-based training group (G_{POWER}), a heart rate-based training group (G_{HEART}), or a control group (G_{CONTROL}). Two subjects were unwilling to participate in the G_{CONTROL} group because of concerns about possible reductions in training status. These subjects were subsequently randomized to either the G_{POWER} or G_{HEART} group.

Subjects

Twenty-one well-trained men cyclists (13) (mean $\pm$ SD ; age = 31 ± 6 years; stature = 1.82 ± 0.07 m; mass = 74.9 ± 8.8 kg) volunteered for this study. After being fully informed of the risks and stresses associated with the study, all subjects completed a Physical Activity Readiness Questionnaire (4), had their training logs analyzed, and had a personal interview about their cycling history. All subjects gave their written informed consent to participate in the study. The experimental protocol was approved by the ethics and research committee of the Faculty of Health Sciences of the University of Cape Town.

Procedures

Each subject completed a 40-km familiarization time trial (TT), a peak aerobic capacity test ($\dot{V}O_{2\text{max}}$), and a repeat 40-km TT test at 7 days, 3 days, and 1 day, respectively, before the start of the 28-day training period.

Subjects reported to the laboratory, which had stable climatic conditions ($22.4 \pm 1.4^\circ\text{C}$, $53 \pm 5\%$ relative humidity, 100.5 ± 8 kPa), at the same time of day for all tests and training sessions. During each performance trial, subjects were blinded to all cues other than completed distance to prevent the adoption of a pacing strategy or performance bias.

Subjects were asked not to eat for at least 2 hours before each of the performance tests or supervised training sessions. Each subject was asked to refrain from training for 24 hours before the $\dot{V}O_{2\text{max}}$ test and to perform a 90-minute submaximal recovery ride (at an intensity below the "lactate turnpoint" determined during $\dot{V}O_{2\text{max}}$ testing [26]) 24 hours before the 40-km TT familiarization and 40-km TT tests. They also were asked to refrain from consuming any caffeine for at least 3 hours before each performance test or training session. Before each testing session, subjects were questioned to confirm that they had adhered to these instructions.

After the 28-day training period, subjects completed a 10-day washout period of training, during which time they were asked not to participate in any racing or prolonged or high-intensity exercise. After the washout period, the $\dot{V}O_{2\text{max}}$ test, 40-km TT test, and anthropometric measurements were

repeated. Subjects were verbally questioned before the testing procedure to ensure that they had adhered to the training protocol.

To confirm that subjects adhered to the prescribed training intensity, each subject was asked to keep a detailed training diary and to record all heart rate data during all training sessions performed outside of the laboratory.

Body Composition Measurements

Stature, body mass, and sum of 7 skinfolds (triceps, biceps, suprailiac, subscapular, calf, thigh, and abdomen) were measured immediately before the 40-km TT test according to the methods described by Ross and Marfell-Jones (24). In addition, body mass was measured before each testing and training session.

Apparatus and Testing Procedure

All testing and training was performed on an electronically braked cycle ergometer (Computrainer Pro 3D, RacerMate, Seattle, Wash). The subject's bicycle was attached to the ergometer by the rear axle quick release mechanism and supported under the front wheel by a plastic support. Rear tire pressure was inflated to 800 kPa before calibration. Calibration and load generator contact pressure was adjusted until the calibration value measured between 0.88 and 0.93 kg. The average calibration value recorded during the study was 0.914 ± 0.02 kg, as recommended by Davison et al. (9,10). Each subject completed a self-paced warm-up for 15 minutes before each testing and training session.

The $\dot{V}O_{2\text{max}}$ and W_{max} tests were performed at a starting workload of $2.5 \text{ W}\cdot\text{kg}^{-1}$ body mass. Load was increased incrementally at a rate of 20 W every minute until the subject could not sustain a cadence of greater than 70 rpm or was volitionally exhausted. During the progressive exercise test, ventilation volume ($\dot{V}_E$), $\dot{V}O_2$, and $\dot{V}CO_2$ were measured for 15-second intervals using an online breath-by-breath gas analyzer and pneumotach (Oxycon, Viasis, Hoechberg, Germany). Calibration of this device was performed during the 15-minute self-paced warm-up according to the manufacturer's instructions. $\dot{V}O_{2\text{max}}$ was recorded as the highest $\dot{V}O_2$ reading recorded for 30 seconds during the test. Peak power output was calculated by averaging the power output for the final minute of the $\dot{V}O_{2\text{max}}$ test.

Subjects completed the self paced maximal 40-km TT on a simulated flat 40-km course. Subjects were allowed to consume water ad libitum throughout the test and were asked to produce the fastest possible 40-km TT time.

Training Protocol

After the testing period, each subject reported to the laboratory on 8 occasions for a supervised training session (2 sessions each week for 4 weeks). Subjects performed the training sessions on their own bicycles mounted to a Computrainer ergometer. After a 15-minute self-paced warm-up period, each subject completed a high-intensity interval training session, consisting of 8 intervals of 4 minutes'

duration, interspersed with 90-second self-paced recovery periods.

Each G_{CONTROL} subject completed a 40-km self-paced training ride twice each week at an intensity below 70% W_{max} (determined during the progressive exercise test). In addition, G_{CONTROL} subjects were asked to perform the same training as the HIT groups outside of the laboratory.

Subjects in the G_{HEART} group completed each interval at a fixed workload corresponding to 80% W_{max} , and subjects in the G_{HEART} group were asked to increase and then maintain a target heart rate equivalent to the heart rate recorded at 80% W_{max} . These values were chosen with the aim of ensuring that both HIT groups trained at the same intensity. Because of the slow half-life of heart rate response (2), subjects in the G_{HEART} group were asked to achieve the target heart rate within 3 minutes during the first interval, within 2 minutes during the second interval, and within 1 minute for all subsequent intervals.

Between each laboratory training session, subjects were asked to adhere to the following training protocol:

–rest day

–90-minute recovery session (at an intensity below the “lactate turnpoint” determined during $\dot{V}O_{2\text{max}}$ testing)

or

–rest day

–90-minute recovery session

–rest day

Data Analyses

Analysis of training data (power output and heart rate) were performed using Cyclingpeaks analysis software, WKO+ edition, version 2.1, 2006 (Peakware Inc., Lafayette, Colo). Data from each training session were filtered using the “Fast Find” function. “Leading Edge” and “Trailing Edge” values as well as minimum and maximum duration values were assigned according to each subject’s absolute training intensity. Filtered data subsequently were inspected visually and corrected before analysis.

Statistical Analyses

Data were analyzed for statistical significance using STATISTICA version 7.0 (Stat-soft Inc., Tulsa, Okla). A 2-way analysis of variance (ANOVA) with repeated measures was used to examine differences in performance measures between the groups before and after HIT intervention or control training. Where a significant difference was found ($p \leq 0.05$) for either the main effect of group or the interaction, a Tukey post hoc analysis was performed. A significant interaction (group $\times$ time)

was interpreted as meaning that the groups responded differently during the training period for that variable. A 1-way ANOVA was performed for the percentage change in each group for all the performance variables. All data are expressed as mean $\pm$ SD.

Further analysis of changes in performance and physiological variables after training were assessed using magnitude-based inferences following the procedure described by Batterham and Hopkins (5). Mean effects of training and their 90% confidence intervals (CIs) were estimated using an Excel spreadsheet (www.sportsci.org/jour/05/ambwgh.htm) with values obtained from a t -test for each independent variable between groups. The spreadsheet computes the chance that the true effect is substantial when a value for the smallest worthwhile change is entered. A value of 1% was defined as a meaningful difference for the performance measures, as used in previous cycling studies (8,21,22). Although we acknowledge the controversy regarding multiple comparisons, we have not tried to correct for any bias using a Bonferroni-adjusted p value because the advantages of this procedure may be outweighed by the disadvantages of this adjustment (23).

RESULTS

One cyclist from G_{POWER} fractured his wrist in a cycling accident, and 1 cyclist from G_{HEART} contracted a viral illness; therefore, their data were excluded from further analysis. On analysis, 2 subjects (1 from G_{HEART} and 1 from G_{POWER}) were found to have pretraining $\dot{V}O_{2\text{max}}$ scores that fell outside 2 SDs from the group average. The data from these subjects were therefore excluded from further analysis. Subsequently, 6 cyclists remained in each experimental group (G_{POWER} and G_{HEART}), and 5 cyclists were in the G_{CONTROL} group. The sample size criteria of 5 subjects per group were met by all groups. Descriptive measures of the cyclists in the 3 groups before the start of the study are shown in Table 1. There were

TABLE 1. Subject characteristics of each group— G_{CONTROL} ($n = 5$), G_{POWER} ($n = 6$), and G_{HEART} ($n = 6$)—expressed as mean $\pm$ SD.

	G_{POWER}	G_{HEART}	G_{CONTROL}
Age (y)	30 $\pm$ 8	30 $\pm$ 5	34 $\pm$ 4
Stature (m)	182 $\pm$ 7	175 $\pm$ 6	182 $\pm$ 3
Body mass (kg)	73 $\pm$ 8	70 $\pm$ 7	82 $\pm$ 9
W_{max} (W)	372 $\pm$ 34	370 $\pm$ 26	369 $\pm$ 46
Relative peak power ($\text{W} \cdot \text{kg}^{-1}$)	5.1 $\pm$ 0.6	5.3 $\pm$ 0.3	4.6 $\pm$ 0.5
$\dot{V}O_{2\text{max}}$ ($\text{ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$)	60 $\pm$ 7	60 $\pm$ 4	54 $\pm$ 7
40-km time trial performance (min:s)	65:14 $\pm$ 2:31	66:15 $\pm$ 2:06	67:22 $\pm$ 1:37
Body fat (%)	13 $\pm$ 3	15 $\pm$ 4	16 $\pm$ 4
Sum of 7 skinfolds (mm)	51 $\pm$ 8	64 $\pm$ 22	72 $\pm$ 26

TABLE 2. Performance characteristics before and after 4 weeks of high-intensity training using either power (G_{POWER} ; $n = 6$), heart rate (G_{HEART} ; $n = 6$), or submaximal training (G_{CONTROL} ; $n = 5$), expressed as mean $\pm$ SD.

	Pre	Post	% Change
$\dot{V}O_{2\text{max}}$ ($\text{ml}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$)			
G_{POWER}	59.9 $\pm$ 7	60.2 $\pm$ 0	+1.06
G_{HEART}	60.3 $\pm$ 4	62.5 $\pm$ 2	+3.75
G_{CONTROL}	54.4 $\pm$ 7	54.4 $\pm$ 7	+0.11
Wmax ($\text{W}\cdot\text{kg}^{-1}$)			
G_{POWER}	5.1 $\pm$ 0.6	5.3 $\pm$ 0.6*	+3.5†
G_{HEART}	5.3 $\pm$ 0.3	5.5 $\pm$ 0.4*	+5.0‡
G_{CONTROL}	4.5 $\pm$ 0.5	4.5 $\pm$ 0.4*	-1.0
40-km time trial time (min:s)			
G_{POWER}	65:14 $\pm$ 2:31	63:43 $\pm$ 1:59§	+2.3
G_{HEART}	66:15 $\pm$ 2:06	64:48 $\pm$ 2:07§	+2.1¶
G_{CONTROL}	67:22 $\pm$ 1:37	67:18 $\pm$ 1:51§	+0.1

Wmax: *Group $\times$ time ($p = 0.01$); † G_{POWER} vs. G_{CONTROL} ($p = 0.05$); ‡ G_{HEART} vs. G_{CONTROL} ($p = 0.01$).

40-km time trial: §Group $\times$ time ($p = 0.01$); || G_{POWER} vs. G_{CONTROL} ($p = 0.006$); ¶ G_{HEART} vs. G_{CONTROL} ($p = 0.01$).

no statistical differences in any of these measurements between groups.

Performance Parameters

The subjects' $\dot{V}O_{2\text{max}}$, Wmax, and 40-km TT performance are shown in Table 2. There were no significant differences in $\dot{V}O_{2\text{max}}$ between groups either before or after training (Figure 1). An ANOVA with repeated measures for Wmax

showed a significant interaction between groups over time ($p < 0.05$; Figure 1). On further analysis, the HIT groups (G_{POWER} and G_{HEART}) increased Wmax scores significantly by 3.5 and 5.0%, respectively, relative to the G_{CONTROL} group ($p < 0.05$; Table 2). There was a significant interaction between groups over time for 40-km TT performance ($p < 0.05$; Figure 1). Relative improvement in 40-km TT performance was significant for both of the HIT groups (G_{POWER} and G_{HEART}) relative to the G_{CONTROL} group, with improvements of 2.3 and 2.1%, respectively ($p < 0.05$; Table 2). There was no significant interaction between the 2 HIT groups for any of the dependent variables.

The mean changes in physiological and performance variables for all groups and magnitude-based statistics for the differences in the changes are shown in Table 3. For $\dot{V}O_{2\text{max}}$ scores, only the G_{HEART} group showed a statistical likelihood of a beneficial effect from training. There were clear-cut beneficial effects on Wmax and 40-km TT performance for both the HIT groups (G_{POWER} and G_{HEART}), with the greatest probability of a beneficial effect being demonstrated for the G_{HEART} group in all of the parameters. For both $\dot{V}O_{2\text{max}}$ and Wmax, there was a statistically “likely” benefit of G_{HEART} over G_{POWER} for improvements in these parameters. In addition, there was a significant correlation ($r = 0.70$; 95% CI: 0.36–0.87) between changes in Wmax and 40-km TT performance for all groups combined (Figure 2).

Training

The objective of prescribing identical average training intensities for each HIT group was achieved during this study. Average power outputs during each interval for groups G_{POWER} and G_{HEART} expressed as percentages of Wmax were 79 ± 1 and $78 \pm 3\%$ Wmax,

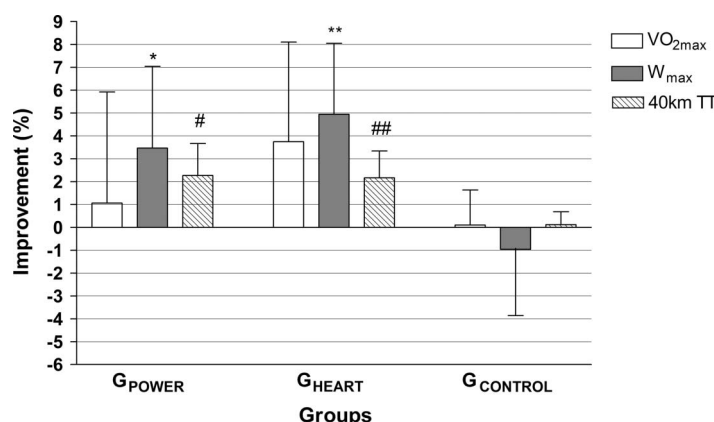


Figure 1. Change in $\dot{V}O_{2\text{max}}$, relative peak workload (Wmax), and 40-km time trial performance (40-km TT) after 4 weeks of high-intensity training based on power (G_{POWER}), heart rate (G_{HEART}), or control (G_{CONTROL}). Wmax: * G_{POWER} vs. G_{CONTROL} ($p = 0.05$); ** G_{HEART} vs. G_{CONTROL} ($p = 0.01$). 40-km TT: # G_{POWER} vs. G_{CONTROL} ($p = 0.006$); ## G_{HEART} vs. G_{CONTROL} ($p = 0.01$; $n = 17$).

TABLE 3. Mean changes in performance and physiological variables after 4 weeks of high-intensity training using either power (G_{POWER} ; $n = 6$), heart rate (G_{HEART} ; $n = 6$), or submaximal training (G_{CONTROL} ; $n = 5$), and chances that the true differences in the changes are substantial.

	Percentage (%) change				Chance that the difference is truly substantial*	
	G_{POWER}	G_{HEART}	G_{CONTROL}	Difference; $\pm$ 90% CI	%	Qualitative
$\dot{V}O_{2\text{max}}$						
G_{POWER} vs. G_{CONTROL}	1.1	—	0.1	1.0; $\pm$ 4.2	49	Possibly
G_{HEART} vs. G_{CONTROL}	—	3.75	0.1	3.6; $\pm$ 3.8	88	Likely
G_{POWER} vs. G_{HEART}	1.1	3.75	—	2.7; $\pm$ 4.8	73	Likely
W_{max}						
G_{POWER} vs. G_{CONTROL}	3.5	—	−1.0	4.5; $\pm$ 3.7	94	Likely
G_{HEART} vs. G_{CONTROL}	—	5.0	−1.0	6.0; $\pm$ 3.3	99	Almost certain
G_{POWER} vs. G_{HEART}	3.5	5.0	—	1.5; $\pm$ 3.5	60	Likely
40-km time trial						
G_{POWER} vs. G_{CONTROL}	2.3	—	0.1	2.2; $\pm$ 1.2	94	Likely
G_{HEART} vs. G_{CONTROL}	—	2.1	0.1	2.0; $\pm$ 1.0	95	Very likely
G_{POWER} vs. G_{HEART}	2.3	2.1	—	0.1; $\pm$ 1.4	13	Unlikely

*Substantial difference = change of $>1.0\%$ for all measures of performance; $\pm$ 90% CL: add and subtract this number to the mean effect to obtain the 90% confidence limits for the true difference.

respectively. The average training intensity of the G_{POWER} and G_{HEART} groups did not differ significantly. This is in accordance with the similar relative heart rate intensity ($G_{\text{POWER}} = 88 \pm 3\%$ HRmax; $G_{\text{HEART}} = 88 \pm 2\%$ HRmax). However, the power profile during intervals was markedly different between these groups. Specifically, the average

5-second maximum power output recorded for each interval differed significantly ($p < 0.001$) between the G_{POWER} and G_{HEART} groups. The average 5-second maximum power output for all intervals for G_{HEART} was 498 ± 89 W, whereas the average 5-second maximum power output for all intervals for G_{POWER} was 401 ± 63 W. Representative examples of

2 randomly chosen subjects are shown in Figure 3. Although the average power outputs and heart rates did not differ, intervals prescribed using heart rate were characterized by a high starting power output followed by a marked decline in power, which is sustained toward the end of the interval. Heart rate increased throughout these intervals until the target heart rate was achieved, after which heart rate remained stable. During power-based intervals, power remained relatively constant throughout, as dictated by the prescribed method, whereas heart rate increased continuously throughout each interval period, with a plateau toward the end of the interval (Figure 3).

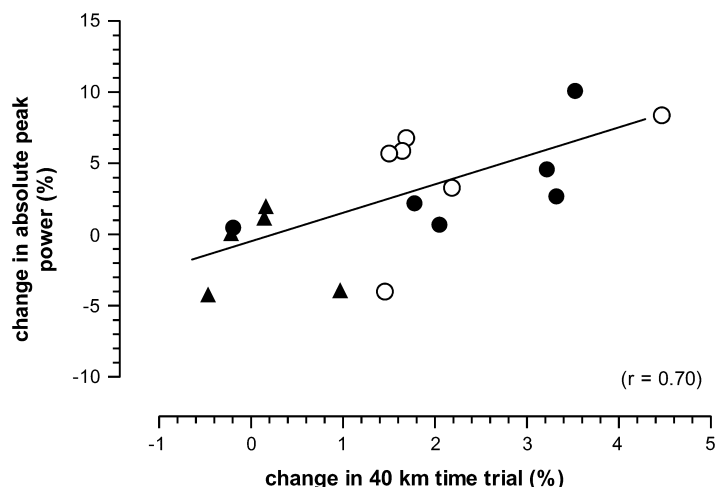


Figure 2. Percentage change in 40-km time trial (40-km TT) performance vs. percentage change in peak workload (W_{max}) after 4 weeks of high-intensity training based on power (G_{POWER} , ●), heart rate (G_{HEART} , ○), or control (G_{CONTROL} , ▲; $n = 17$).

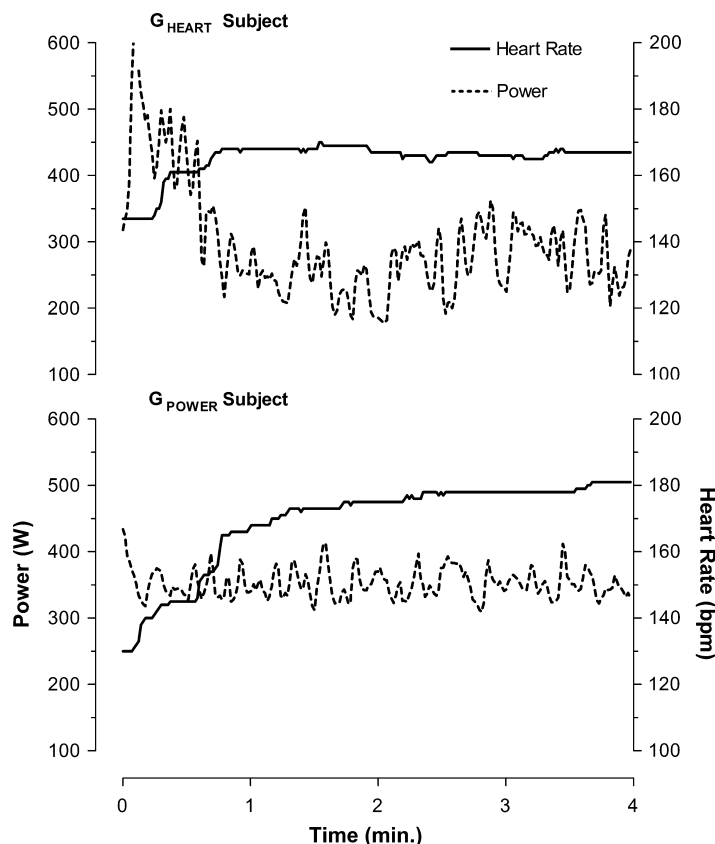


Figure 3. Representative examples of changes in power and heart rate during a single interval of 2 arbitrary chosen subjects using either heart rate (G_{HEART} ; a) or power (G_{POWER} ; b) to prescribe training ($n = 17$).

DISCUSSION

The present study has shown that HIT performed at either a fixed heart rate or fixed power elicited improvements in W_{max} and 40-km TT performance in already well-trained cyclists. The improvements in W_{max} (range 3.5–5.0%) and 40-km TT performance (range 2.1–2.3%) were similar to changes previously reported using similar HIT protocols and conducted over similar time periods (11).

Although we cannot state unequivocally that, for this specific HIT protocol, heart rate-based training elicited greater improvements in performance and related parameters than power-based training, there is evidence in our data to suggest that this may be so. Analyses inferring about magnitudes (5) show that there is a “likely” beneficial effect of heart rate-based training in comparison with power-based training, particularly with reference to improvements in W_{max} and $\dot{V}O_{2\text{max}}$ (Figure 1). Given that the performances of elite athletes are separated by small margins that are indiscernible and that cannot be detected with null-hypothesis testing (5,8), it is appropriate to infer about the magnitude of the differences between groups. This approach

suggests that there may be advantages in using heart rate compared with power output prescribed training for this specific protocol. However, further research is needed to confirm these findings.

The average power output (79 ± 1 vs. $78 \pm 3\%$ W_{max} for G_{POWER} vs. G_{HEART}) and heart rate (88 ± 3 vs. $88 \pm 2\%$ HR_{max} for G_{POWER} vs. G_{HEART}) during training intervals did not differ between the 2 HIT groups. However, the G_{HEART} group had a different power profile during intervals, which may, therefore, have resulted in short periods of training above $\dot{V}O_{2\text{max}}$ and W_{max} values (supramaximal training; Figure 3). Previous studies have shown that training at or above $\dot{V}O_{2\text{max}}$ intensity may be the most effective means of eliciting additional improvements in already highly trained athletes (6,7,14,18,25).

Further research is needed to compare responses to heart rate- and power-based training in the field and to compare the training effects at differing workloads and interval durations.

In addition, a comparison of training responses using either fixed or variable power outputs within intervals may provide more valuable data for the optimal use of power meters to prescribe training. Power-based training may be more beneficial for training protocols such as T-max intervals (16,18), but evidence for other power-based sessions is still lacking.

In summary, HIT prescribed using either heart rate or power improved performance (W_{max} and 40-km TT) similarly. However, magnitude-based inferences suggest that there may be an advantage of using target heart rate over power output for this particular HIT protocol.

PRACTICAL APPLICATIONS

The prescription of training based on heart rate has the potential to be inaccurate if the conditions under which the training is prescribed are not controlled. Although power is a more direct method of prescribing training, a potential limitation of this method is that power output training zones change significantly over a relatively short time period. This necessitates frequent maximal testing to update these values.

In contrast, heart rate-based training zones remain relatively stable throughout prolonged periods of time, which makes this method easier and less dependent on frequent testing than power-based training. In conclusion, although power meters have proven to be accurate and useful tools to monitor training, exercise prescription by power has not proven to have additional benefits over heart rate when prescribing HIT. When coaches are unable to monitor training progress frequently, we recommend prescription using heart rate-based training. Coaches also should consider that heart rate-based intervals performed under stable conditions may provide an additional advantage over power-based intervals.

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